

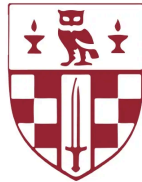
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A machine learning approach to predicting ICU admission in traumatic brain injury patients using emergency department clinical data

Research project proposal submitted in partial fulfillment of requirements for MSc Data Science and Artificial Intelligence to be awarded 2027

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Declaration

I, *Anna Daisy FitzPatrick*, declare that the work carried out and reported on by this document and the contents of this document, titled '*A machine learning approach to predicting ICU admission in traumatic brain injury patients using emergency department clinical data*', and accompanying artefacts, except for where attributed to other sources by way of citations within the body of this document and listed under the References section of this document, is formed of my own original work, which has never before been produced for any other purpose, and has been created solely as part of delivery of *MSc Data Science and AI research project proposal* for or under supervision of *Dr Roman Kontchakov*.

All information in this document and accompanying artefacts has been obtained and presented in accordance with the relevant ethical conduct and academic rules.

Signature:

Anna Daisy FitzPatrick

Date:

11/06/2026

Contents

1. Introduction	5
1.1 Traumatic Brain Injuries as a Clinical Burden	5
1.2 Machine Learning for Clinical Decision Support	5
1.3 Project Rationale	5
2. Aims and Objectives	5
3. Literature Review	6
3.1 Current Approaches to TBI Triage and ICU Admission	6
3.2 Machine Learning for Clinical Prediction in Emergency Medicine	6
3.3 Machine Learning Applications in TBI	7
3.4 Clinical Databases in Machine Learning Research	8
3.5 Explainable AI in Healthcare	8
3.6 Research Gap and Justification	9
4. Data Sources	9
5. Methodology	11
5.1 Data Preparation and Preprocessing	11
5.2 Model Development	12
5.3 Evaluation Framework	12
5.4 Interpretability Analysis	13
6. Project Timeline	13
7. Ethical Considerations	13
8. Risk Register	14
9. References	15

Abstract

Traumatic brain injury (TBI) is a major emergency department (ED) presentation and a leading cause of intensive care unit (ICU) admission, yet decisions at triage rely on subjective clinical judgement. This project develops and evaluates supervised machine learning (ML) models to predict ICU admission to enable earlier intervention and resource allocation. Using only triage-available data extracted from MIMIC-IV-ED and linked to outcomes in MIMIC-IV, multiple classification algorithms - logistic regression, random forest, and gradient boosting - will be trained and assessed using AUROC, recall, F1-score and calibration metrics, with interpretability methods identifying the most influential triage variables. Findings aim to inform the development of real-time decision support tools for TBI triage.

1. Introduction

1.1 TBI as a Clinical Burden

TBI encompasses a spectrum from mild concussion to devastating structural damage, and is a leading driver of critical care admission and long-term disability. Its heterogeneity - where a subset of apparently mild presentations deteriorate rapidly - means accurate triage-level risk stratification is clinically critical (Maas et al., 2022), with the ICU stratum representing 47% of patients (Steyerberg et al., 2019). Despite this burden, triage continues to rely on subjective clinical judgement susceptible to variability and error.

1.2 Machine Learning in Clinical Decision Support

ML-based clinical prediction has expanded substantially alongside large-scale de-identified electronic health record (EHR) databases, with demonstrated success across imaging, cardiology, and emergency medicine (Fernandes et al., 2020; Sánchez-Salmerón et al., 2021). In TBI specifically, approaches range from logistic regression - the clinical benchmark - to ensemble methods, including XGBoost, which have demonstrated superior discriminative power on structured clinical data (Wang et al., 2022).

1.3 Project Rationale

Despite this evidence base, existing ML studies applied to TBI outcomes predominantly rely on features unavailable at ED presentation, including CT findings, laboratory values, or assessments collected hours after admission. Prediction of ICU admission has received little targeted attention. This distinction matters clinically: delayed or missed ICU admission in TBI exposes patients to secondary injury through unmonitored intracranial pressure and hypotension (Farahvar et al., 2012). Accordingly, this project addresses a specific, evidence-defined gap: developing a model that predicts ICU admission using only variables available at ED triage.

2. Aims and Objectives

The aim of this project is to evaluate whether supervised ML models trained on ED data can predict ICU admission in TBI patients, and to assess their utility and limitations as support tools for triage.

The objectives are to:

1. **Critically review** literature on TBI triage, ML in emergency medicine, and EHR-based predictive modelling to identify methodological gaps
2. **Examine** the suitability of MIMIC-IV-ED data for predicting ICU admission, including preprocessing, feature selection, class imbalance handling, and missing data management
3. **Develop** and **critically evaluate** supervised ML classifiers - logistic regression, random forest, and gradient boosting - using cross-validation, discriminative, and calibrative metrics
4. **Analyse** feature importance and clinical relevance using SHAP analysis to identify key drivers of ICU admission prediction

5. **Discuss** the generalisability, ethical considerations, and deployment constraints of the developed models

3. Literature Review

3.1 Current Approaches to TBI Triage and ICU Admission

Triage is standardised through protocols such as the Emergency Severity Index (ESI), though its evaluation in the literature has focused primarily on short-term mortality rather than ICU admission (Najafi et al., 2018). Within this framework, TBI triage relies primarily on the Glasgow Coma Scale (GCS), shown in Figure 1. Despite its ubiquity since 1974 (Teasdale & Jennett, 1974), inter-rater reliability is particularly poor in TBI patients (Zhang et al., 2025), and GCS demonstrates a ceiling effect that limits utility for ICU admission decisions: among patients presenting with GCS-15, 30% required hospital admission, of whom 29% were admitted to ICU - yet no GCS-derived variables significantly predicted ICU admission in multivariate analysis (Ganti et al., 2019), directly motivating the case for data-driven decision support.

Table 10-2: Glasgow Coma Scale

Eye Opening		Verbal Response		Motor Response	
	Points		Points		Points
Spontaneous	4	Oriented	5	Obeys commands	6
To voice	3	Confused	4	Localizes pain	5
To pain	2	Inappropriate words	3	Withdraws	4
None	1	Incomprehensible sounds	2	Abnormal flexion	3*
		Silent	1	Abnormal extension	2**
				No movement	1

* Decorticate posturing to pain.
 ** Decerebrate posturing to pain.

Figure 1: Glasgow Coma Scale, scored from 3 to 15, used to assess consciousness in TBI patients. (Campbell, 2016)

Beyond GCS, a significant proportion of patients are under- or over-triaged, and no validated, TBI-specific triage tools have been identified in the literature (Alqurashi et al., 2022). The principal challenge is not the absence of clinical indicators, but the difficulty of consistently integrating multiple imperfect predictors under time-constrained conditions.

3.2 Machine Learning for Clinical Prediction in Emergency Medicine

ML methods offer theoretical advantage over rule-based scoring systems with complex and non-linear clinical relationships. A systematic review by Sánchez-Salmerón et al. (2021) found ML models demonstrated superior performance for mortality, hospitalisation, and critical care outcomes compared with conventional triage scales, with tree-based ensemble methods frequently achieving the strongest discriminative performance. Across the literature, evidence suggests that triage-available variables contain substantial predictive signal for downstream clinical outcomes. Studies using vital signs, chief complaint, and arrival characteristics have consistently demonstrated that ML models can predict admission and critical care requirements with performance exceeding conventional triage approaches and,

in some cases, experienced clinicians (Fernandes et al., 2020; Kachman et al., 2025). An example of this performance is shown in Figure 2.

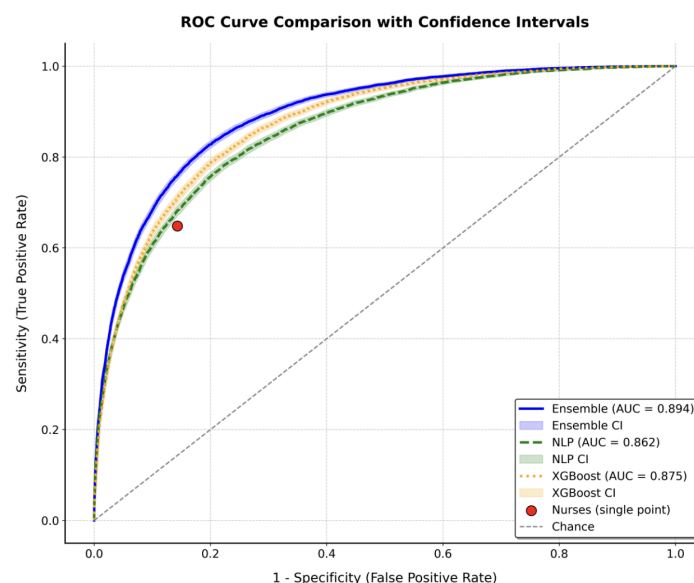


Figure 5: ROC curves comparing the ensemble model (AUC = 0.894), NLP (AUC = 0.862), and XGBoost (AUC = 0.875) with the performance point of nurses.

Figure 2: ROC curves from Kachman et al. (2025) shows ML models outperforming experienced triage nurses for hospital admission prediction

Persistent limitations exist, however. Generalisability is a primary concern; a systematic review found that whilst seven studies demonstrated rigorous methodology with AUROCs of 0.81-0.93, the remainder carried high risk of bias (Williams et al., 2025). Reliance on AUROC alone can overstate clinical utility in class-imbalanced settings, making recall and F1-score particularly important where missed cases carry significant risk. A further limitation is incorporation of temporally downstream variables unavailable at triage, risking information leakage that inflates reported performance. Whether these limitations are compounded in TBI specifically is examined in Section 3.3.

3.3 Machine Learning Applications in TBI

TBI outcome prediction has historically been dominated by the IMPACT and CRASH prognostic models, which remain the clinical benchmark (Maas et al., 2007; MRC CRASH Trial Collaborators, 2008) but their reliance on admission-level CT and clinical variables limits applicability at triage. Subsequent ML studies have largely focused on in-hospital mortality, with Wu et al. (2023) finding significant heterogeneity and high risk of bias across 14 studies, reflecting the absence of a standardised methodological framework, displayed in Figure 3. Importantly, the framework highlights substantial methodological heterogeneity across studies, reinforcing concerns regarding reproducibility and helping explain why comparative evaluation of TBI prediction models remains difficult.

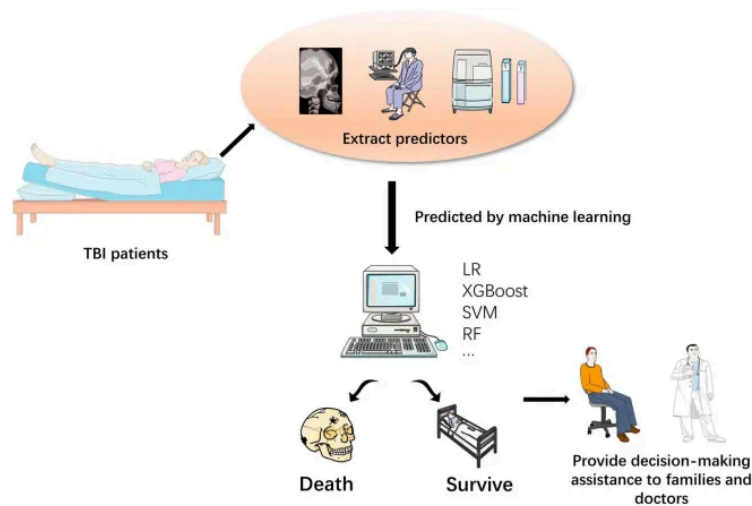


Figure 3: Framework from Wu et al. (2023) shows the flow from patient data to TBI mortality prediction across 14 studies

Critically, ML superiority over traditional approaches cannot be assumed in TBI. Across TBI-specific studies, evidence regarding the superiority of complex ML models remains mixed. Whilst ensemble methods often outperform traditional approaches in heterogeneous emergency medicine populations, several studies suggest that these advantages diminish in lower-dimensional TBI datasets, where logistic regression frequently remains competitive (Gravesteijn et al., 2020). Validation practices also remain poor - five of 14 studies reviewed by Wu et al. (2023) performed no validation whatsoever, with robustness and reproducibility identified as the weakest methodological domains across 39 AI-based TBI studies (Malhotra et al., 2025). The predominant focus on mortality over ICU admission as a triage-relevant outcome represents a further underaddressed gap.

3.4 Clinical Databases in Machine Learning Research

MIMIC has emerged as the most widely adopted platform for critical care ML research (Johnson et al., 2023), though its use carries limitations directly relevant to this project. As a single-centre US academic dataset, findings may not generalise across healthcare systems, a concern compounded by documented racial disparities in ED triage where Black and Hispanic patients are often assigned lower acuity scores for subjective presentations including neurological complaints (Peitzman et al., 2023). The absence of standardised preprocessing pipelines also leads to inconsistent cohort construction across nominally identical tasks (Gupta et al., 2022). Despite wide use of MIMIC-IV for ICU prediction, the linked MIMIC-IV-ED module remains underexploited for ED-specific research - the ED-to-ICU linkage it enables represents both a gap in the literature and the methodological foundation.

3.5 Explainable AI in Healthcare

Complex ensemble methods frequently achieve strong discriminative performance, but their opacity poses a barrier to clinical adoption - clinicians cannot be expected to act on risk scores without understanding the reasoning behind them (Pierce et al., 2022). SHAP (SHapley Additive exPlanations) has become the dominant interpretability framework in response, assigning each feature a contribution value for individual predictions based on cooperative game theory (Lundberg & Lee, 2017), and providing both local and global explanations with consistency guarantees. Limitations remain - additive explanations can

mislead in correlated feature spaces, and “explainable AI” (XAI) solutions have often failed to address real-world clinician workflow needs (Khene et al., 2025).

In ED triage, interpretability is a prerequisite for adoption rather than an optional enhancement. A model predicting ICU admission that cannot identify which triage variables drove a given prediction offers limited value to a time-pressured clinician (Kim et al., 2025), directly motivating the inclusion of SHAP analysis as a clinical plausibility requirement in this project. Taken together, the literature suggests that clinical utility depends not solely on predictive performance, but on achieving an appropriate balance between discrimination, calibration, transparency, and methodological robustness.

3.6 Research Gap and Justification

Three themes emerge from this review. First, existing ML models for TBI routinely incorporate variables unavailable at triage, restricting real-time applicability - addressed here by restricting all predictors to triage-available variables. Second, inconsistent validation methodology undermines confidence in reported performance - addressed through strict train-test separation, within-fold SMOTE, and TRIPOD+AI-compliant reporting. Third, the systematic focus on mortality over ICU admission reflects a mis-match between what models predict and what triage clinicians need to decide; ICU admission is a proxy outcome influenced by system-level factors that include bed availability and clinician decision-making, requiring careful interpretation that this project explicitly acknowledges. No studies were identified that develop and evaluate an interpretable ML model predicting ICU admission in TBI patients using exclusively triage-available data.

4. Data Sources

MIMIC-IV-ED is a publicly available database of approximately 425,000 ED stays at Beth Israel Deaconess Medical Center (2011-2019), organised around a central *edstays* table linked to *diagnosis*, *medrecon*, *pyxis*, *triage*, and *vitalsign* tables shown in Figures 4 and 5. The *triage* table captures variables recorded at initial assessment, including temperature, heart rate, and pain scores; the *diagnosis* table provides ICD-10 codes used to identify the TBI cohort (S06.x). The variables shown in Figure 4 are recorded during initial assessment and therefore align with the project's objective of developing a model that could realistically operate at the point of triage without reliance on downstream clinical information. In Figure 5, the linkage structure between tables is illustrated. It enables separation of predictor and outcome sources, allowing triage variables to be extracted from MIMIC-IV-ED whilst ICU admission labels are obtained independently from MIMIC-IV, reducing the risk of information leakage. An important note is that injury codes are retrospectively assigned rather than recorded at triage, which will be acknowledged as a limitation in cohort construction.

subject_id	stay_id	temperature	heartrate	resprate	o2sat	sbp	dbp	pain	acuity	chiefcomplaint
10000032	32952584	97.8000	87.0000	14.0000	97.0000	71.0000	43.0000	7	2.0000	Hypotension
10000032	33258284	98.4000	70.0000	16.0000	97.0000	106.0000	63.0000	0	3.0000	Abd pain, Abdominal distention
10000032	35968195	99.4000	105.0000	18.0000	96.0000	106.0000	57.0000	10	3.0000	n/v/d, Abd pain
10000032	38112554	98.9000	88.0000	18.0000	97.0000	116.0000	88.0000	10	3.0000	Abdominal distention
10000032	39399961	98.7000	77.0000	16.0000	98.0000	96.0000	50.0000	13	2.0000	Abdominal distention, Abd pain, LETHAGIC
10000084	35203156	97.5000	78.0000	16.0000	100.0000	114.0000	71.0000	0	2.0000	Confusion, Hallucinations
10000084	36954971	98.7000	80.0000	16.0000	95.0000	111.0000	72.0000	0	2.0000	Altered mental status, B Pedal edema
10000108	32522732	98.2100	83.0000	20.0000	100.0000	112.0000	81.0000	5	3.0000	L CHEEK ABSCESS
10000108	36533795	98.8000	98.0000	16.0000	100.0000	135.0000	85.0000	5	3.0000	LEFT CHEEK SWELLING, Abscess
10000108	39513268	98.8000	101.0000	18.0000	100.0000	131.0000	62.0000	5	3.0000	L FACIAL SWELLING

Figure 4: Sample of unfiltered MIMIC-IV-ED triage table, illustrating triage-available variables.

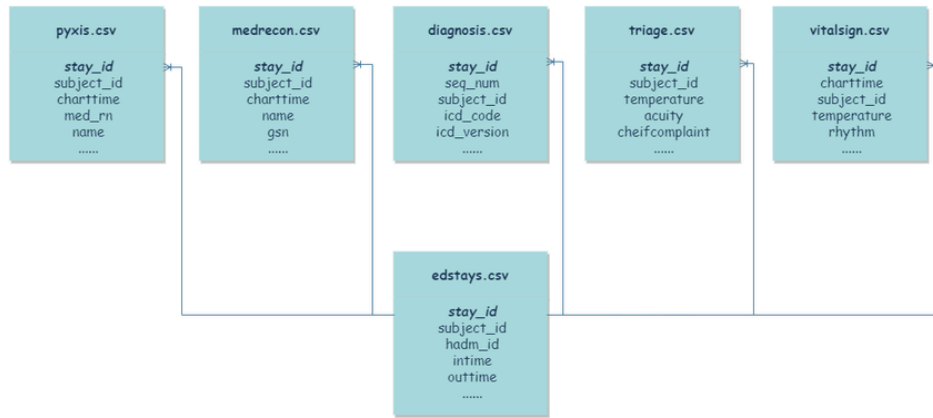


Figure 5: Entity-relationship diagram of the MIMIC-IV-ED dataset (Wei et al., 2025), illustrating the central `edstays` table links to its subtables via `stay_id`, with linkage to MIMIC-IV enabled through `subject_id` and `hadm_id`.

MIMIC-IV supplies the outcome label only. Its `icustays` table determines whether a linked admission included an ICU stay, constituting the binary outcome variable. Post-admission monitoring data is excluded as it accounts for leakage. The two databases are linked via shared patient-level identifiers: `subject_id` and `hadm_id` - an architecture underexploited in the literature. MIMIC-IV-ED was selected over alternatives including eICU database, which lacks an ED triage module (Pollard et al., 2018), and NHAMCS, which does not support patient-level linkage to ICU outcomes (CDC, 2022).

5. Methodology

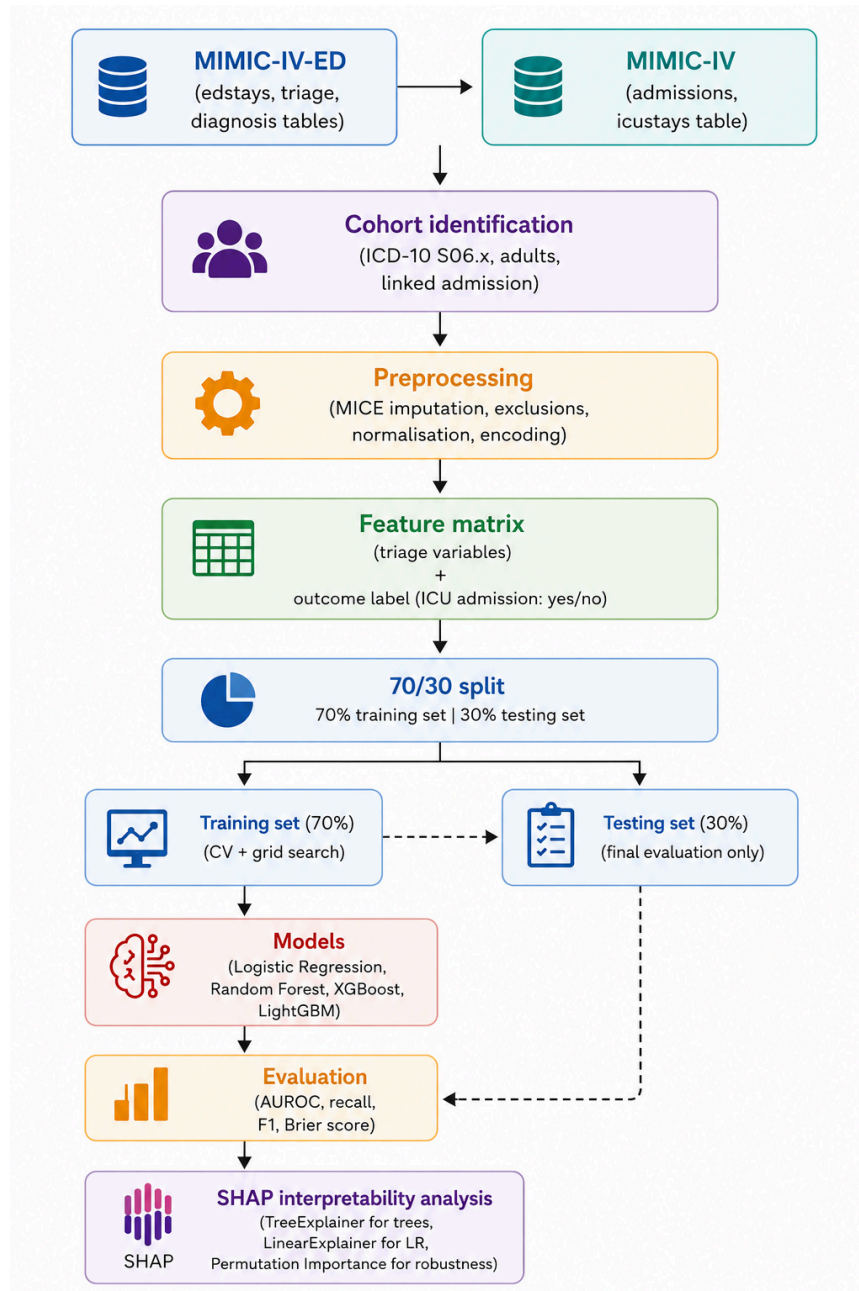


Figure 6: The pipeline reflects the project's central methodological principle of maintaining strict temporal and procedural separation between training and testing data, thus minimising information leakage and supporting unbiased performance estimation.

5.1 Data Preparation and Preprocessing

The TBI cohort will be identified by filtering MIMIC-IV-ED diagnosis records for ICD-10 TBI codes (S06.x), restricted to adult patients with a linked MIMIC-IV admission record. All predictor variables are drawn exclusively from the *triage* table, ensuring the model reflects information realistically available at ED presentation.

Missing vital signs will be handled using multiple imputation via chained equations (MICE), which outperforms complete case analysis and mean imputation in clinical prediction contexts (Sisk et al., 2023), with patients exceeding a pre-specified missingness threshold excluded. An important note is that MICE assumes data is missing at random (MAR) - this is

an assumption that cannot be verified directly, and may not hold in MIMIC triage data where missingness in vital signs is plausibly related to a patient's severity of injury. Continuous features will be z-score normalised and categorical variables one-hot encoded. Correlation thresholding will reduce multicollinearity prior to training, with permutation importance assessed post-hoc to evaluate predictor contribution (Fisher et al., 2019). Class imbalance will be addressed using SMOTE applied within each training fold and class weight adjustment, evaluated using recall and F1-score rather than AUROC alone (Chawla et al., 2002). If synthetic oversampling leads to overfitting or degraded calibration, cost-sensitive learning and decision threshold optimisation will be explored as fallbacks.

5.2 Model Development

Three supervised classifiers will be developed and compared: logistic regression, random forest, and gradient boosting, reflecting a progression from an interpretable baseline to increasingly complex ensemble methods. Logistic regression serves as the primary baseline. Christodoulou et al. (2019) found no consistent performance benefit of ML over logistic regression in clinical prediction tasks, making it an essential comparator against which added complexity must be justified.

Random forest (Breiman, 2001) captures nonlinear interactions, while gradient boosting (Friedman, 2001) will be evaluated via XGBoost and LightGBM implementations, both compatible with SHAP-based interpretability. SVMs were excluded due to scalability limitations under class imbalance and reduced SHAP compatibility. Should gradient boosting exhibit overfitting despite regularisation, logistic regression will be retained as the preferred deployable model. Neural networks were excluded as complexity does not reliably improve performance in low-dimensional settings (Gravesteijn et al., 2020). The dataset will be partitioned 70/30, stratified by outcome. All cross-validation and hyperparameter tuning via grid search will be conducted exclusively on the training set using 5-fold cross validation, with the testing set reserved solely for final evaluation to eliminate leakage.

5.3 Evaluation Framework

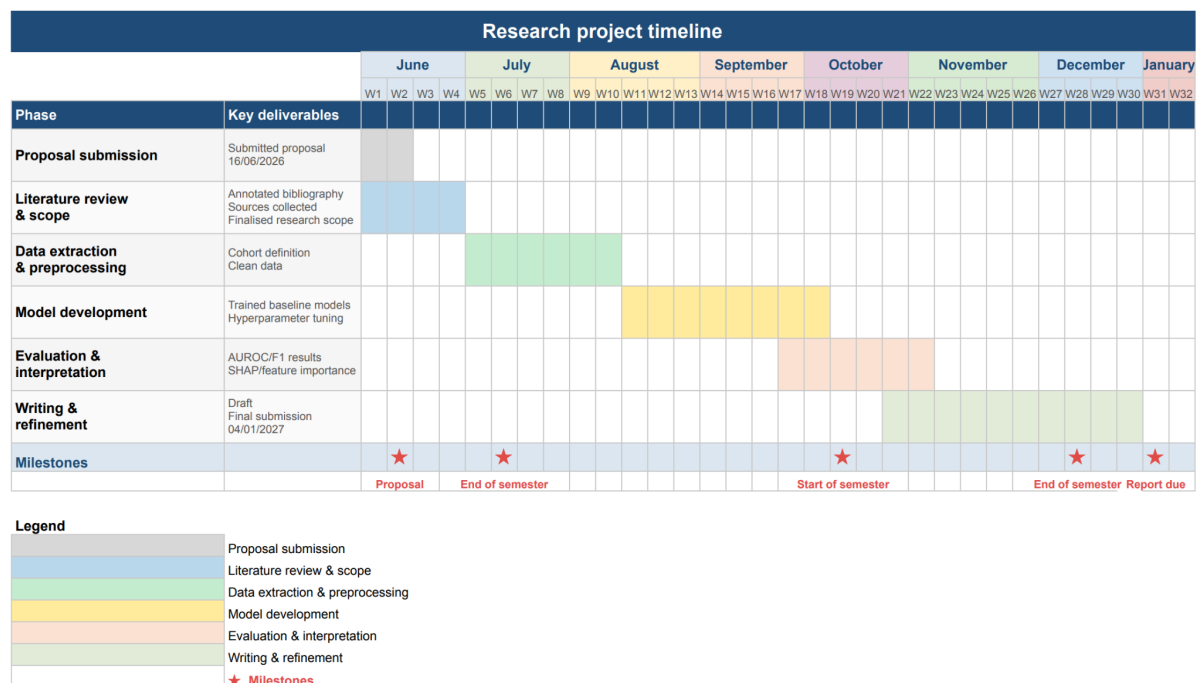
Models will be evaluated using AUROC, recall, F1-score, and Brier score. Recall is prioritised as the clinically critical metric - missing a TBI patient requiring ICU admission carries greater consequence than a false positive - while the Brier score assesses probabilistic calibration, essential where risk scores must be clinically meaningful rather than merely discriminative (Van Calster et al., 2019). Precision-recall curves will also be examined to assess performance under class imbalance. No subgroup-level fairness analysis will be conducted, representing a further limitation. External validation is beyond scope; internal stratified cross-validation provides robustness but cannot fully assess cross-institutional reproducibility. Reporting will follow TRIPOD+ AI guidelines (Collins et al., 2024).

5.4 Interpretability Analysis

SHAP will be applied to all trained models to quantify feature contributions to predictions (Lundberg & Lee, 2017), aggregated across the dataset to generate global measures of feature importance. TreeSHAP will be used for tree-based models and LinearSHAP for logistic regression. Where multicollinearity affects interpretability, permutation importance will be used as a sensitivity analysis alongside SHAP rankings (Fisher et al., 2019). SHAP-identified predictors will be compared against established clinical indicators including GCS, systolic blood pressure, respiratory rate, and age (Steyerberg et al., 2008; Maas et al., 2022) to assess clinical plausibility of model behaviour.

6. Project Timeline

The project spans 30 weeks across five overlapping phases, from literature review and data access through to model development, evaluation, and final write-up, with a two-week contingency buffer built into the final phase.



8. Risk register

Risk	Likelihood	Impact	Mitigation
Cohort size too small after pre-processing or linkage failure	Medium	Lack of statistical power	Broaden ICD inclusion criteria or relax missing data thresholds
Class imbalance (ICU vs non-ICU)	High	Biased classifier optimising for majority class	Apply class-weight adjustments, prioritise recall and F1 over accuracy
Model captures historical triage biases	Medium	Reduced fairness and generalisability	Interpret importance carefully, acknowledge dataset limits
Overfitting	Medium	Poor generalisation to unseen data	k-fold cross-validation, regularisation, constrained tree depth
ICD-10 coding inconsistency	Medium	Miscoded or missing diagnoses may bias cohort	Validate cohort against expected distribution, inclusion criteria transparently
Project timeline slippage	Low	Compression of later phases and write up	Build two-week buffer into pre-processing, define minimum viable pipeline early

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